STATISTICAL PROGRAMMING LAB	
Course Code: ADL47	**Credits: 0:0:1**
Pre – requisites: Basic Programming	**Contact Hours: 14P**
Course Coordinator: Dr. Sowmya B J	

Course Contents

The students are required to

1. Understand and Interpret the basic concepts, Syntaxes in R
2. Demonstrate the vector data objects operations
3. Implement the concept of matrix, array and factors
4. Demonstrate the usage of data frames in statistical analysis.
5. Demonstrate the creation of sample (Dummy) Data and perform data manipulation
6. Study and implementation of various control structures in R
7. Demonstrate data manipulation operation with dplyr and data.table package
8. Study and implementation of Data Visualization with ggplot2
9. Study and implementation data transpose operations in Statistical analysis.
10. Demonstrate Descriptive statistics in R
11. Implementing Statistical operations, T-test, Paired Test,
12. Implementing Statistical analysis like correlation, Chi Square test, Analysis of Variance and Correlation.

Suggested Learning Resources

Reference Books:

1. An Introduction to Statistical Programming Methods with R, Matthew Beckman, Stéphane Guerrier, Justin Lee, Roberto Molinari, Samuel Orso & Iegor Rudnytskyi, https://smac-group.github.io/ds/, 2020
2. Norman Matloff, The Art of R Programming, UC Davis 2009.
3. Michael Akritas, " Probability & Statistics with R for Engineers and Scientists‖, 2nd Edition on, CRC Press, 2016.
4. Crawley, M. J. (2006), ―Statistics - An introduction using R‖, John Wiley, London 32.
5. ―Statistical Programming in R‖, Oxford University Press, 2016

Course Outcomes (COs):

At the end of the course the student will be able to:

1. Demonstrate the basic concepts of R, use of Data frames and various control structures (PO1,2,3,4,512, PSO1,2 and 3)
2. Illustrate Data Manipulation and Visualization (PO1,2,3,4,5, 12, PSO1,2 and 3)
3. Appraise different statistical operations. (PO1,2,3,4,5,12, PSO1,2 and 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	**Marks**	**Course Outcomes (COs) addressed**
Lab Test	20	CO1, CO2, CO3
Weekly Evaluation-Lab Record	30	-
The Final CIE out of 50 Marks = Marks of Lab Record + Marks scored in Lab Test		
Semester End Examination (SEE)		
Course End Examination (One full question from the Lab Question Bank, Programs will be coded using C and executed)	50	CO1, CO2, CO3